

AAIC 2026 · BASICS OF GENETIC ANALYSIS

Genome-Wide Association Study (GWAS)

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ISTAART immersive workshop · Session 4



SESSION 4 / GWAS

What this session covers

01

Refresher

SNP Analysis

02

Multiple testing

From one test to millions

03

Running a GWAS with PLINK

File formats and the association command

04

Reading the output

Plotting, quality checks, and functional follow-up

01 / REFRESHER

SNP analysis

A single-SNP test is one regression at one locus

Linear (continuous outcome) or logistic (case / control)

SNP is coded additively as 0, 1 or 2 — the number of minor alleles a person carries

Outcome = Intercept + **SNP** + covariates

Covariates typically include age, sex and ancestry principal components



02 / MULTIPLE TESTING

From one test to millions

GWAS repeat the single-SNP test across millions of variants

Testing that many hypotheses inflates false positives, so the significance threshold must be far stricter than 0.05

Alternatives to a fixed Bonferroni line include FDR control and permutation testing

$$5 \times 10^{-8}$$

Genome-wide significance threshold

Derived from an assumed ~1 million independent tests across the genome.

03

Running a GWAS with PLINK

File formats, the association command, and good defaults

PLINK 1.9 • pre-installed on the workshop VM

03 / RUNNING A GWAS WITH PLINK

The files we work with

GENETIC FILES

.bed	binary genotype data (compressed)
.bim	variant information (text)
.fam	sample information (text)

Best practice: only modify these with PLINK itself

INPUT FILES (CAN BE COMBINED)

- Outcome / phenotype
- Covariates
- Numeric data only — encode categories as dummy variables
- Missing data flagged as -9 by default

03 / RUNNING A GWAS WITH PLINK

The association command

```
./plink \  
  --bfile ADNI1.impQC.rs \  
  --adjust \  
  --linear hide-covar \  
  --covar demographics.txt \  
  --covar-name AGE,SEX,C1-C4 \  
  --pheno demographics.txt \  
  --pheno-name Whole_hippocampus \  
  --maf 0.02 --pfilter 1 \  
  --out ADNI1_HV_GWAS
```

■ Specific to your project

files and column names

■ Recommended filters

--maf drops rare variants; --pfilter trims output

■ Optional flags

--adjust adds corrected p-values; hide-covar tidies output

Change --linear for --logistic when the outcome is a case / control trait

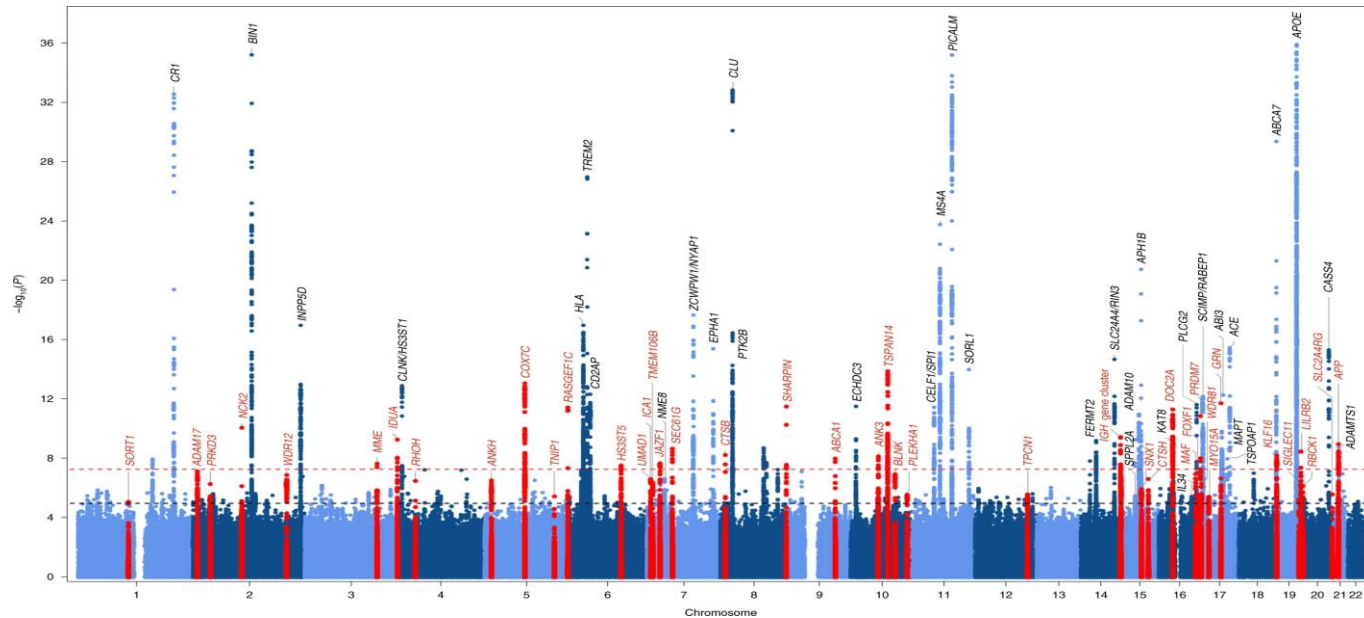
04

Reading the output

Plotting, quality checks, and functional follow-up

04 / READING THE OUTPUT

The Manhattan plot

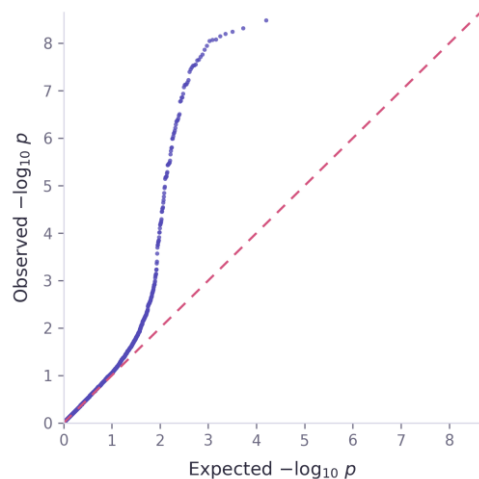


Each dot is a SNP — x-axis genomic location, y-axis $-\log_{10} p$. Red dotted line represents the genome-wide significance level ($P = 5 \times 10^{-8}$), and the black dotted line represents the suggestive significance level ($P = 1 \times 10^{-5}$)

Adapted from Bellenguez et al., 2022

04 / READING THE OUTPUT

The QQ plot



Observed p-values plotted against expected by chance

Points should follow the diagonal, lifting away only in the upper tail where **true signal** sits

An early, whole-line departure suggests population stratification or other confounding — revisit QC and covariates

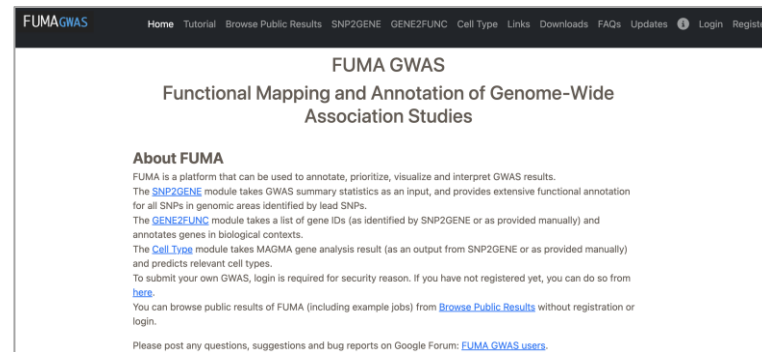
04 / READING THE OUTPUT

Functional follow-up with FUMA

Annotation, prioritization and visualization of GWAS results

- **SNP2GENE:** from variants to gene annotation
- **GENE2FUNC:** from genes to biological annotation
- **MAGMA:** gene-set analysis

Tools such as LDSC complement this with heritability and genetic-correlation estimates



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Key takeaways

- 01 A GWAS is the single-SNP test scaled to the whole genome
- 02 Strict thresholds (5×10^{-8}) keep false positives in check
- 03 PLINK 1.9 runs the association; sensible filters matter
- 04 Manhattan and QQ plots show running mistakes; functional annotation is crucial

THANK YOU

Questions?

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